

H524 Introduction to Biostatistics

Week 7: Nonparametric Methods

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What We've Learned So Far

Parametric Methods (Weeks 1-6):

- One-sample and two-sample t-tests
- Paired t-test
- ANOVA for comparing multiple groups
- All assume approximately normal data

Key Assumptions:

- Data approximately normally distributed
- Large enough sample size
- Equal variances (for some tests)

Question: What if these assumptions are violated?

When Parametric Assumptions Fail

Common situations:

- **Skewed data:** Income, hospital stays, survival times
- **Outliers:** Extreme values that distort means
- **Ordinal data:** Pain scales (1-10), satisfaction ratings
- **Small samples:** $n < 20$, normality uncertain

Example: Hospital Stay Data

5 patients: 2, 3, 4, 3, 45 days

- Mean = 11.4 days (distorted by outlier)
- Median = 3 days (more representative)
- t-test assumes normality (violated here)

Solution: Nonparametric tests

What Are Nonparametric Tests?

Key idea: Use **ranks** instead of actual values

Advantages:

- No normality assumption
- Robust to outliers
- Work with ordinal data
- Valid for small samples

Trade-off:

- Slightly less powerful than parametric when data ARE normal
- But safer when assumptions violated

When in doubt, use nonparametric!

Ranking Example

Original data: 3, 10, 5, 8, 2

Ranks (smallest to largest):

- 2 → Rank 1 (smallest)
- 3 → Rank 2
- 5 → Rank 3
- 8 → Rank 4
- 10 → Rank 5 (largest)

Why use ranks?

- Outliers become just "highest rank"
- Ordinal data already in rank form
- No distribution assumptions needed

Tied Ranks: What If Values Are Equal?

Problem: What if two observations have the same value?

Solution: Assign average of the ranks they would occupy

Example: Pain scores: 2, 4, 4, 6, 7

Value	Position	Tied?	Final Rank
2	1	No	1
4	2 & 3	Yes	$(2 + 3)/2 = 2.5$
4	2 & 3	Yes	$(2 + 3)/2 = 2.5$
6	4	No	4
7	5	No	5

R does this automatically! No manual calculation needed.

Nonparametric Tests We'll Learn

Parametric Test	Nonparametric Alternative
One-sample t-test	Sign Test Wilcoxon Signed-Rank
Paired t-test	Sign Test Wilcoxon Signed-Rank
Two-sample t-test	Wilcoxon Rank-Sum (Mann-Whitney)
One-way ANOVA	Kruskal-Wallis

Focus today: How to run each test in R and interpret results

Sign Test: The Simplest Test

When to use:

- One sample (test median) OR
- Paired data (before/after)
- Data not normal OR ordinal scale

How it works:

- 1 Calculate differences from hypothesized value
- 2 Count positive (+) vs negative (-) signs
- 3 If median = test value, expect 50/50 split
- 4 Use binomial test for p-value

Ignores: Magnitude of differences (only direction)

Advantage: Works with any data type

R function: `binom.test()`

Sign Test Example 1: One Sample

Question: Is median recovery time different from 6 days?

Data: 8, 5, 7, 10, 4, 9, 3, 12, 6

```
# Data
recovery <- c(8, 5, 7, 10, 4, 9, 3, 12, 6)

# Differences from 6
diff <- recovery - 6

# Count positive (ignoring zeros)
n_positive <- sum(diff > 0) # 5
n_total <- sum(diff != 0)    # 8

# Sign test
binom.test(n_positive, n_total, p=0.5)
```

Output: p-value = 0.727

Interpretation: No significant difference ($p > 0.05$)

Sign Test Example 2: Paired Data

Question: Does treatment reduce pain scores?

```
# Pain scores (1-10 scale)
before <- c(7, 8, 6, 9, 5, 8, 7, 6)
after <- c(5, 6, 4, 7, 4, 6, 5, 5)

# Differences
diff <- before - after
# All positive means pain decreased

# Count signs
n_improved <- sum(diff > 0)   # 8 improved
n_total <- sum(diff != 0)     # 8 total

# Sign test
binom.test(n_improved, n_total, p=0.5)
```

Output: p-value = 0.008

Interpretation: $p < 0.05$, treatment DOES reduce pain significantly

Sign Test: Interpretation

Hypotheses:

- H_0 : Median = hypothesized value (or no difference)
- H_A : Median $\neq$ hypothesized value (two-sided)

Decision:

- If $p < 0.05$: Reject H_0 , significant difference
- If $p \geq 0.05$: Fail to reject H_0 , no significant difference

Limitations:

- Only uses direction (+ or -), not magnitude
- Less powerful than tests that use magnitude
- But works with ANY type of data (even severe outliers)

Sign Test Example 3: One-Sided Test

Question: Does a new therapy reduce recovery time below 14 days?

Data: Recovery times for 10 patients

```
recovery <- c(12, 15, 10, 13, 16, 11, 14, 9, 12, 15)
```

```
# Count how many are below 14
```

```
below_14 <- sum(recovery < 14)
```

```
total <- length(recovery)
```

```
# One-sided test: H1: median < 14
```

```
binom.test(below_14, total, p=0.5,  
            alternative="greater")
```

Output: p-value = 0.377

Interpretation: $p \geq 0.05$, NOT enough evidence that median $\neq$ 14 days

Sign Test Example 4: Comparing Proportions

Question: Do patients prefer treatment A over B?

Data: 20 patients tried both, 15 preferred A, 5 preferred B

```
# Test if preference differs from 50/50  
prefer_A <- 15  
total <- 20
```

```
binom.test(prefer_A, total, p=0.5)
```

Output: p-value = 0.041

Interpretation: $p < 0.05$, patients DO significantly prefer treatment A

Note: Sign test is flexible - works for preferences, ordinal ratings, any +
/- data

Wilcoxon: More Powerful Than Sign Test

When to use:

- Same situations as sign test
- One sample OR paired data
- Data not normal

Difference from sign test:

- Sign test: Only counts direction (+ or -)
- Wilcoxon: Uses BOTH direction AND magnitude
- Ranks absolute differences, then applies signs

Result: More powerful (better at detecting differences)

R function: `wilcox.test()`

Wilcoxon: How Magnitude is Incorporated

Simple example - Before/After treatment:

Patient	Difference	Sign	Rank by Size
A	-1	-	1 (smallest)
B	+2	+	2
C	+8	+	3 (largest)

Sign test: 1 negative, 2 positive (ignores that +8 is huge!)

Wilcoxon:

- Rank by size: 1, 2, 8 $\rightarrow$ ranks 1, 2, 3
- Apply signs: -1, +2, +3
- Test statistic uses signed ranks
- Large differences get more weight!

Key insight: Wilcoxon gives more credit to larger changes

Wilcoxon Example 1: One Sample

Same question: Is median recovery time different from 6 days?

Same data: 8, 5, 7, 10, 4, 9, 3, 12, 6

```
# Data
```

```
recovery <- c(8, 5, 7, 10, 4, 9, 3, 12, 6)
```

```
# Wilcoxon signed-rank test
```

```
wilcox.test(recovery, mu=6)
```

Output: p-value = 0.325

Compare:

- Sign test: $p = 0.727$
- Wilcoxon: $p = 0.325$ (smaller, more powerful)
- Both say no significant difference ($p > 0.05$)

Wilcoxon Example 2: Paired Data

Question: Does medication reduce blood pressure?

```
# Blood pressure before and after medication
bp_before <- c(145, 138, 150, 142, 156, 148)
bp_after  <- c(138, 135, 142, 140, 148, 145)

# Wilcoxon paired test
wilcox.test(bp_before, bp_after, paired=TRUE)

# Summary statistics
mean(bp_before)  # 146.5
mean(bp_after)   # 141.3
mean(bp_before - bp_after)  # 5.2 mmHg reduction
```

Output: $p\text{-value} = 0.035$

Interpretation: $p < 0.05$, medication DOES reduce BP significantly
Average reduction = 5.2 mmHg

Wilcoxon Example 3: Pain Score Study

Question: Does physical therapy reduce pain?

```
# Pain scores (0-10 scale) before and after 8 weeks PT
pain_before <- c(7, 8, 6, 9, 7, 8, 6, 7, 8, 9)
pain_after  <- c(5, 6, 5, 7, 6, 7, 4, 6, 7, 8)

# Wilcoxon test
wilcox.test(pain_before, pain_after, paired=TRUE)

# Effect size
median(pain_before) # 7.5
median(pain_after)  # 6
# Median reduction = 1.5 points
```

Output: p-value = 0.011

Interpretation: PT significantly reduces pain ($p < 0.05$)

Median reduction = 1.5 points on pain scale

Sign Test vs Wilcoxon: When to Use Each?

Feature	Sign Test	Wilcoxon
Uses direction?	Yes	Yes
Uses magnitude?	No	Yes
Power	Lower	Higher
Extreme outliers	Handles well	Handles well
Ordinal data	Perfect	Good
Recommendation	Very robust data	Most situations

In practice: Use Wilcoxon (more powerful, R handles it easily)

Exception: If data are truly ordinal with unequal intervals, sign test safer

Wilcoxon Example 3: Power Advantage

Question: Weight loss after 8-week program

Data: Before-after differences (kg): 1.2, 0.8, 2.5, 0.3, 1.8, 0.5, 1.1

```
diff <- c(1.2, 0.8, 2.5, 0.3, 1.8, 0.5, 1.1)
```

```
# Sign test
```

```
binom.test(sum(diff > 0), length(diff), p=0.5)
```

```
# p-value = 0.016
```

```
# Wilcoxon test
```

```
wilcox.test(diff, mu=0)
```

```
# p-value = 0.016
```

Result: Both significant here. But Wilcoxon usually more powerful with larger differences.

Wilcoxon Example 4: Depression Scores

Question: Does therapy reduce depression (scale 0-50)?

```
before <- c(42, 38, 45, 40, 48, 35, 43)
after  <- c(38, 36, 40, 35, 42, 32, 39)
```

```
# Paired Wilcoxon test
wilcox.test(before, after, paired=TRUE)
```

```
# Summary statistics
mean(before - after) # 5.1 point reduction
median(before - after) # 5 point reduction
```

Output: p-value = 0.022

Interpretation: $p < 0.05$, therapy DOES significantly reduce depression scores

Wilcoxon Example 5: One-Sample with Outlier

Question: Is median cholesterol different from 200?

Data: 185, 195, 210, 198, 425 (outlier!), 192, 205

```
cholesterol <- c(185, 195, 210, 198, 425, 192, 205)
```

```
# Wilcoxon handles outliers well  
wilcox.test(cholesterol, mu=200)
```

```
# Compare to t-test (affected by outlier)  
t.test(cholesterol, mu=200)
```

Wilcoxon: $p = 1.00$, fail to reject H_0

t-test: $p = 0.39$, also fail to reject but more affected by outlier

Advantage: Wilcoxon robust to extreme values

Wilcoxon Rank-Sum: Two Independent Groups

When to use:

- Comparing TWO independent groups
- Data not normal OR small samples OR outliers
- Nonparametric version of two-sample t-test

Also called: Mann-Whitney U test (same test, different name)

How it works:

- Combine both groups and rank all values
- Sum ranks for each group
- If groups equal, rank sums should be similar

R function: `wilcox.test()` (without `paired=TRUE`)

Tests: Do the two groups have different medians?

Wilcoxon Rank-Sum Example 1: Treatment Comparison

Question: Do Treatment A and B differ in response time?

```
# Response times (days)
treatment_A <- c(23, 31, 25, 28, 30)
treatment_B <- c(18, 22, 20, 24, 19)

# Visualize
boxplot(treatment_A, treatment_B,
        names=c("Treatment A", "Treatment B"),
        ylab="Response Time (days)")

# Wilcoxon rank-sum test
wilcox.test(treatment_A, treatment_B)

# Medians
median(treatment_A) # 28
median(treatment_B) # 20
```

Output: p-value = 0.016

Interpretation: Treatment B shorter (20 vs 28 days, $p < 0.05$)

Wilcoxon Rank-Sum Example 2: Drug Study

Question: Do two drugs differ in effectiveness?

```
# Symptom relief scores (higher = better)
drug_X <- c(45, 52, 48, 55, 50, 47, 53)
drug_Y <- c(38, 42, 40, 45, 39, 43, 41)

# Wilcoxon test
wilcox.test(drug_X, drug_Y)

# Summary
mean(drug_X); mean(drug_Y)      # 50 vs 41
median(drug_X); median(drug_Y)  # 50 vs 41
```

Output: $p\text{-value} = 0.003$

Interpretation: Drug X more effective ($p < 0.05$), 9 point improvement

Wilcoxon Rank-Sum Example 3: Hospital Stays

Question: Do two procedures differ in hospital stay length?

```
# Hospital stay (days)
procedure_1 <- c(3, 4, 5, 3, 6, 4, 7, 5, 4, 12)
procedure_2 <- c(5, 6, 8, 7, 9, 6, 8, 10, 7, 15)

# Check for outliers
boxplot(procedure_1, procedure_2,
        names=c("Procedure 1", "Procedure 2"))

# Note: Outliers present (12, 15 days)
# Use nonparametric!

wilcox.test(procedure_1, procedure_2)
```

Output: p-value = 0.011

Interpretation: Procedure 1 has shorter stays ($p < 0.05$)

Median: 4.5 vs 7.5 days

Rank-Sum vs. T-Test: When to Use Which

Use Wilcoxon Rank-Sum when:

- Small sample sizes ($n < 20$ per group)
- Skewed data or outliers present
- Ordinal data (ratings, scales)
- Can't verify normality

Use Two-Sample T-Test when:

- Large samples ($n \geq 30$ per group)
- Data approximately normal
- Want to compare means (not medians)

When in doubt: Use Wilcoxon (safer, minimal power loss)

Kruskal-Wallis: Three or More Groups

When to use:

- Comparing THREE OR MORE independent groups
- Data not normal OR small samples
- Nonparametric version of one-way ANOVA

How it works:

- Rank all data across all groups
- Compare rank sums between groups
- If groups equal, rank sums should be similar

R function: `kruskal.test()`

Tests: Do the groups have different medians?

Important: If significant, need post-hoc tests to see WHICH groups differ

Kruskal-Wallis Example 1: Three Drugs

Question: Do three drugs differ in pain relief?

```
# Pain relief scores (0-10, higher = better)
drug_A <- c(2, 3, 4, 3)
drug_B <- c(4, 6, 5, 7)
drug_C <- c(5, 8, 9, 10)

# Combine data
pain <- c(drug_A, drug_B, drug_C)
drug <- factor(rep(c("A", "B", "C"), each=4))

# Kruskal-Wallis test
kruskal.test(pain ~ drug)

# Medians
tapply(pain, drug, median)
# A: 3, B: 5.5, C: 8.5
```

Output: p-value = 0.016

Interpretation: At least one drug differs ($p < 0.05$)

Why Do We Need Bonferroni Correction?

The problem: Multiple comparisons inflate Type I error

Simple example - 3 groups:

- Need 3 pairwise tests: A vs B, A vs C, B vs C
- Each test has 5% false positive rate ($\alpha = 0.05$)
- Overall chance of at least one false positive: $\sim 14\%$!
- With 4 groups (6 tests): $\sim 26\%$ false positive rate

Bonferroni solution: Make each test stricter

- Divides α by number of tests
- 3 tests: each uses $\alpha = 0.05/3 = 0.017$
- Keeps overall error rate at 5%

In R: `p.adjust.method="bonferroni"` does this automatically

Post-Hoc Tests After Kruskal-Wallis

Kruskal-Wallis tells us: "At least one group differs"

Post-hoc tells us: "Which specific pairs differ?"

```
# Same data from previous slide
# Pairwise comparisons with Bonferroni correction
pairwise.wilcox.test(pain, drug,
                     p.adjust.method="bonferroni")
```

Output p-values:

- A vs B: $p = 0.050$ (borderline)
- A vs C: $p = 0.014$ (significant)
- B vs C: $p = 0.050$ (borderline)

Interpretation: Drug C significantly better than Drug A

Kruskal-Wallis Example 2: Four Diets

Question: Do four diets differ in weight loss?

```
# Weight loss (kg) after 3 months
diet_A <- c(2.1, 3.2, 2.5, 3.0, 2.8)
diet_B <- c(3.5, 4.2, 3.8, 4.0, 3.9)
diet_C <- c(4.5, 5.1, 4.8, 5.3, 4.9)
diet_D <- c(2.8, 3.0, 2.9, 3.1, 2.7)

# Combine
weight_loss <- c(diet_A, diet_B, diet_C, diet_D)
diet <- factor(rep(c("A", "B", "C", "D"), each=5))

# Test
kruskal.test(weight_loss ~ diet)
```

Output: $p\text{-value} < 0.001$

Interpretation: Diets differ significantly

Next step: Post-hoc to see which pairs differ

Kruskal-Wallis Example 3: Treatment Groups

Question: Do 5 treatments differ in recovery time?

```
# Recovery times (days) for 5 treatments
# n = 6 per group
recovery <- c(
  12, 14, 13, 15, 11, 13, # Treatment 1
  10, 11, 9, 12, 10, 11, # Treatment 2
  15, 17, 16, 18, 16, 15, # Treatment 3
  8, 9, 7, 10, 8, 9,      # Treatment 4
  13, 14, 12, 15, 13, 14 # Treatment 5
)
treatment <- factor(rep(1:5, each=6))

kruskal.test(recovery ~ treatment)
```

Output: p-value < 0.001

Interpretation: Treatments differ significantly
Treatment 4 appears fastest (median 8.5 days)

Post-Hoc Comparisons After Kruskal-Wallis

Problem: Kruskal-Wallis tells us groups differ, but NOT which pairs

Solution: Pairwise Wilcoxon with Bonferroni correction

```
# Using sleep medication data from earlier
sleep_hours <- c(5.2, 6.1, 5.8, 6.3, 5.5,    # Low
                 6.8, 7.2, 7.5, 6.9, 7.1,    # Medium
                 7.8, 8.2, 8.5, 7.9, 8.1)    # High
dose <- factor(rep(c("Low","Med","High"), each=5))

# Post-hoc test
pairwise.wilcox.test(sleep_hours, dose,
                     p.adjust.method="bonferroni")
```

Result: All three pairs differ significantly ($p < 0.05$ for each)

Bonferroni: Adjusts for multiple comparisons to control Type I error

Kruskal-Wallis vs. ANOVA: Comparison

Feature	ANOVA	Kruskal-Wallis
Assumption	Normality	None
Compares	Means	Medians
Power (normal data)	Higher	Slightly lower
Power (skewed data)	Lower	Higher
Outliers	Sensitive	Robust
Sample size	Need $n \geq 30$	Works with small n
Post-hoc	Tukey HSD	Pairwise Wilcoxon

Recommendation:

- If data normal: Use ANOVA (more powerful)
- If data skewed/outliers/small n: Use Kruskal-Wallis (safer)

Example: When Kruskal-Wallis Wins

Data: Pain scores (0-10) for 3 treatments, with outliers

```
pain <- c(3, 4, 3, 5, 15,      # Treatment A (outlier!)
          6, 7, 6, 8, 7,      # Treatment B
          2, 3, 2, 4, 3)      # Treatment C
treatment <- factor(rep(c("A","B","C"), each=5))

# ANOVA (affected by outlier)
summary(aov(pain ~ treatment))
# p = 0.13 (not significant)

# Kruskal-Wallis (robust to outlier)
kruskal.test(pain ~ treatment)
# p = 0.01 (significant!)
```

Lesson: Outliers can hide real differences in ANOVA
Kruskal-Wallis detects the pattern correctly

Decision Guide: Step 1 - Check Normality

Always check normality first!

Visual methods:

- Histogram: Should look bell-shaped
- Q-Q plot: Points should follow the line
- Boxplot: Should be symmetric

Statistical test:

- Use `shapiro.test()`
- If $p \geq 0.05$: Normal OK → Use parametric
- If $p < 0.05$: Not normal → Use nonparametric

Rule of thumb: When in doubt, use nonparametric (safer!)

Decision Guide: Step 2 - Choose Your Test

Scenario	Normal?	Test
One sample	Yes	t-test
	No	Wilcoxon
Paired data	Yes	Paired t-test
	No	Wilcoxon
Two groups	Yes	Two-sample t-test
	No	Wilcoxon rank-sum
3+ groups	Yes	ANOVA
	No	Kruskal-Wallis

Remember:

- Check normality **FIRST**
- If normal: More power with parametric
- If not normal: Use nonparametric (robust)

Checking Normality in R

Always check normality first!

```
# Visual check
hist(data)
qqnorm(data)
qqline(data)
boxplot(data)

# Statistical test
shapiro.test(data)

# Interpretation:
# p >= 0.05: Normality assumption reasonable
#           -> Use parametric test
# p < 0.05:  Not normal
#           -> Use nonparametric test
```

Example: If `shapiro.test` gives $p = 0.03$, use Wilcoxon instead of t-test

When to Use Each Nonparametric Test

Sign Test:

- Ordinal data or extreme outliers
- Simplest possible test

Wilcoxon Signed-Rank:

- One sample or paired data
- Continuous but not normal
- More powerful than sign test

Wilcoxon Rank-Sum:

- Two independent groups
- Continuous but not normal
- Alternative to two-sample t-test

Kruskal-Wallis:

- 3+ independent groups
- Continuous but not normal
- Alternative to ANOVA, use post-hoc if significant

Decision Flowchart

Step 1: Check normality (Shapiro-Wilk, Q-Q plot)

If NORMAL: Use parametric test

- One sample: One-sample t-test
- Paired: Paired t-test
- Two independent: Two-sample t-test
- 3+ independent: One-way ANOVA

If NOT NORMAL: Use nonparametric test

- One sample: Sign test or Wilcoxon signed-rank
- Paired: Sign test or Wilcoxon signed-rank
- Two independent: Wilcoxon rank-sum
- 3+ independent: Kruskal-Wallis

When in doubt: Use nonparametric (safer)

Quick Reference: R Functions

Test	R Function
Check normality	<code>shapiro.test(data)</code> <code>qqnorm(data); qqline(data)</code>
Sign test	<code>binom.test(n+, n, 0.5)</code>
Wilcoxon signed-rank (one sample)	<code>wilcox.test(x, mu=...)</code> <code>wilcox.test(x, mu=0)</code>
Wilcoxon signed-rank (paired)	<code>wilcox.test(before, after, paired=TRUE)</code>
Wilcoxon rank-sum	<code>wilcox.test(group1, group2)</code>
Kruskal-Wallis	<code>kruskal.test(y ~ group)</code>
Post-hoc	<code>pairwise.wilcox.test(y, group, p.adjust="bonferroni")</code>

Comparing Parametric vs Nonparametric Results

Example: Blood pressure data

```
bp_before <- c(145, 138, 150, 142, 156, 148)
bp_after <- c(138, 135, 142, 140, 148, 145)

# Check normality
diff <- bp_before - bp_after
shapiro.test(diff) # p = 0.071 (borderline)

# Parametric: Paired t-test
t.test(bp_before, bp_after, paired=TRUE)
# p-value = 0.006

# Nonparametric: Wilcoxon
wilcox.test(bp_before, bp_after, paired=TRUE)
# p-value = 0.035
```

Both significant! t-test slightly more powerful (smaller p)

Power: Parametric vs Nonparametric

When data **ARE** normal:

- Parametric tests have **MORE** power
- More likely to detect real differences
- Smaller p-values for same effect size

When data **NOT** normal:

- Parametric tests may give **WRONG** results
- Inflated Type I error rates
- Nonparametric tests give **CORRECT** results

Trade-off:

- Small power loss with nonparametric if data normal
- But correct results regardless of distribution
- **When in doubt, use nonparametric!**

Common Scenarios and Test Choice

Scenario 1: Small sample ($n=8$), unknown distribution

- **Use nonparametric** (can't verify normality)

Scenario 2: Large sample ($n=100$), slightly skewed

- **Could use parametric** (CLT helps)
- Or use nonparametric to be safe

Scenario 3: Pain scale data (1-10 ordinal)

- **Use nonparametric** (ordinal scale)

Scenario 4: Shapiro $p = 0.60$ (clearly normal)

- **Use parametric** (more power)

Scenario 5: One extreme outlier present

- **Use nonparametric** (robust to outliers)

Power Loss: How Much Do We Sacrifice?

Question: If data ARE normal, how much power do we lose using nonparametric?

Answer: Very little!

Test Comparison	Relative Efficiency
Wilcoxon vs. t-test	95%
Sign test vs. t-test	64%
Kruskal-Wallis vs. ANOVA	95%

Interpretation:

- Wilcoxon/Kruskal-Wallis lose only 5% power vs parametric
- Sign test loses more (36%), but still useful for robustness
- In practice: Use nonparametric when in doubt

Bottom line: Nonparametric tests are remarkably efficient!

Nonparametric Tests in R:

Test	R Function
Sign test	<code>binom.test(n_pos, n_total, p=0.5)</code>
Wilcoxon signed-rank	<code>wilcox.test(data, mu=value)</code>
Wilcoxon paired	<code>wilcox.test(x, y, paired=TRUE)</code>
Wilcoxon rank-sum	<code>wilcox.test(group1, group2)</code>
Kruskal-Wallis	<code>kruskal.test(outcome ~ group)</code>
Post-hoc tests	<code>pairwise.wilcox.test()</code>
Check normality	<code>shapiro.test(data)</code>

Interpretation: Always the same

- If $p < 0.05$: Significant difference
- If $p \geq 0.05$: No significant difference

When to Use Nonparametric Tests

Use nonparametric when:

- Data significantly skewed (Shapiro $p < 0.05$)
- Extreme outliers present
- Small sample size ($n < 20$)
- Ordinal data (pain scales, ratings)
- Can't verify normality assumption

Advantages:

- Robust (not affected by outliers)
- No distribution assumptions
- Valid for small samples

Trade-off:

- Slightly less powerful when data IS normal
- But safer when unsure!

- 1 **Always check normality first** (`shapiro.test()`)
- 2 **Choose parametric when normal, nonparametric when not**
- 3 **Nonparametric tests are easy in R:**
 - `wilcox.test()` for most situations
 - `kruskal.test()` for 3+ groups
- 4 **Interpretation same as always:**
 - $p < 0.05$ = significant
 - $p \geq 0.05$ = not significant
- 5 **When in doubt, use nonparametric (safer!)**

Common Mistakes to Avoid

Mistake 1: Forgetting to check normality

- Always run `shapiro.test()` before choosing test

Mistake 2: Using parametric test with small n and non-normal data

- CLT doesn't save you with $n < 20$ and skewed data

Mistake 3: Forgetting `paired=TRUE` for paired data

- `wilcox.test(before, after)` is WRONG for paired data
- `wilcox.test(before, after, paired=TRUE)` is CORRECT

Mistake 4: Not doing post-hoc after Kruskal-Wallis

- Kruskal-Wallis only says "groups differ"
- Need `pairwise.wilcox.test()` to see which pairs

Mistake 5: Overthinking it

- When in doubt, use nonparametric. It's almost always fine!

Practice in Lab:

- Check normality with visual and statistical methods
- Run each type of nonparametric test
- Compare parametric vs. nonparametric results
- Choose appropriate test for different scenarios

Worksheet:

- Apply these tests to real data
- Practice interpretation
- Understand when each test is appropriate

Remember: Focus on WHEN to use each test and HOW to interpret results!